

Assignment No. 2 (Solution hint+key)

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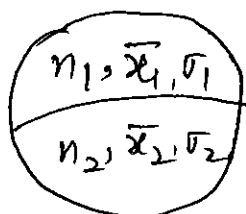
1. $n_1 = 200, n_2 = 150, n_3 = 250$
 $\bar{x}_1 = 25.5, \bar{x}_2 = 31.0, \bar{x}_3 = 42.5$
 $\sigma_1 = 7.0, \sigma_2 = 5.0, \sigma_3 = 8.0$

Combined mean $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3}{n_1 + n_2 + n_3}$

Combined S.D. $\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + n_3(\sigma_3^2 + d_3^2)}{(n_1 + n_2 + n_3)}}$

$d_1 = \bar{x}_1 - \bar{x}, d_2 = \bar{x}_2 - \bar{x}, d_3 = \bar{x}_3 - \bar{x}$

2. $n = 100, \bar{x} = 8, \sigma = \sqrt{10.5}$



$n_1 = 50, \bar{x}_1 = 10, \sigma_1 = 2$

$n_2 = 50, \bar{x}_2 = ?, \sigma_2 = ?$

Combined mean $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{(n_1 + n_2)}$ from this formula, we will get $\bar{x}_2$.

Combined S.D. $\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)}}$ (20)

using $\bar{x}_2$, from (20) we will get σ_2 .

3. $n = 100, \bar{x} = 60, \sigma = 10$

wrong observations: 5, 4.5

correct observations: 30, 20

$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i' \Rightarrow \sum_{i=1}^n x_i' = n\bar{x} = 100 \times 60 = 6000$
 this is wrong sum

Correct $\sum_{i=1}^n x_i' = 6000 - 5 - 4.5 + 30 + 20 = ()$

Similar $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$
 $\sum_{i=1}^n x_i^2 = n(\sigma^2 + \bar{x}^2)$

$$\begin{aligned} \text{Hence Correct } \sum_{i=1}^n x_i^2 &= n(\sigma^2 + \bar{x}^2) - (5)^2 - (4.5)^2 + (30)^2 + (20)^2 \\ &= 100((10)^2 + (60)^2) - (5)^2 - (4.5)^2 + (30)^2 + (20)^2 \\ &= () \end{aligned}$$

$$\text{Hence Correct Mean} = \frac{\text{Correct } \sum_{i=1}^n x_i}{100} = \bar{x}^* (\text{say})$$

$$\text{Correct S.D. } \sigma = \sqrt{\frac{1}{n} \left(\text{Correct } \sum_{i=1}^n x_i^2 \right) - \bar{x}^{*2}} = (\sigma^*) \text{ say}$$

4.

$$n = 20, \bar{x} = 10, \sigma = 2$$

one item 8 was incorrect.

1) Mean + S.D. when wrong item is omitted,

$$\text{Sol}^n: \text{modified/Correct } \sum x_i' = n\bar{x} - 8 = 20 \times 10 - 8 = 192$$

$$\text{Correct Mean} = \frac{192}{19} = 10.11$$

$$\begin{aligned} \text{modified/Correct } \sum x_i'^2 &= n(\bar{x}^2 + \sigma^2) - (8)^2 \\ &= 20(100 + 4) - 64 = 2016 \end{aligned}$$

$$\begin{aligned} \text{Correct S.D. } (\sigma) &= \sqrt{\frac{1}{n} \sum x_i'^2 - \bar{x}^2} \\ &= \sqrt{\frac{1}{19}(2016) - (10.11)^2} = \sqrt{\frac{106.11 - 102.21}{19}} \\ &= \sqrt{\frac{3.9}{19}} = 1.97 \end{aligned}$$

11) If it is replaced by 12

$$\text{Correct } \sum_{i=1}^n x_i' = n\bar{x} - 8 + 12 = 200 - 8 + 12 = 204$$

$$\text{Correct Mean} = \frac{204}{20} = 10.2$$

$$\text{Correct } \sum_{i=1}^n x_i'^2 = n(\bar{x}^2 + \sigma^2) - 8^2 + 12^2 = 2080 - 64 + 144 = 2160$$

$$\begin{aligned} \text{Correct S.D. } (\sigma) &= \sqrt{\frac{1}{n} \sum x_i'^2 - \bar{x}^2} = \sqrt{\frac{2160}{20} - (10.2)^2} = \sqrt{108 - 104.04} \\ &= \sqrt{3.96} = 1.99 \end{aligned}$$

5. $n = 10, \bar{x} = 9.5 + \sigma = 2.5$

additional observation 15.0

$$\sum_{i=1}^{10} x_i = 10 \times 9.5 = 95$$

Sum of 11 observations $\sum_{i=1}^{11} x_i = 95 + 15.0 = 110$

Mean of 11 observations $\bar{x} = \frac{110}{11} = \underline{\underline{10}}$

$$\sum_{i=1}^{10} x_i^2 = 10(\bar{x}^2 + \sigma^2) = 10(90.25 + 6.25) = 10 \times 96.5 = 965$$

Hence sum of squares of 11 observations is

$$\sum_{i=1}^{11} x_i^2 = 965 + (15.0)^2 = 965 + 225 = 1190$$

S.D. of 11 observations is

$$\sigma = \sqrt{\frac{1}{11}(1190) - (10)^2} = \sqrt{108.18 - 100} = \sqrt{8.18} = \underline{\underline{2.86}}$$

6.

$$n = 10, V = \frac{X-30}{2}, \Sigma V = -1, \Sigma V^2 = 49$$

Correct observation 40, Wrong observation 30,

Hence Correct $\Sigma V_c = -1 + \frac{40-30}{2} - \frac{30-30}{2}$
 $= -1 + 5 = 4$

Correct Mean $\bar{x} = A + h\bar{V} = 30 + 2\left(\frac{1}{10} \times 4\right)$
 $= 30.8$

Correct $\Sigma V_c^2 = 49 + \left(\frac{40-30}{2}\right)^2 - \left(\frac{30-30}{2}\right)^2$
 $= 49 + 25 = 74$

Correct $\sigma_x^2 = h^2 \sigma_{V_c}^2 = 4 \times \left(\frac{1}{n} \Sigma V_c^2 - \bar{V}_c^2\right)$
 $= 4 \times \left(\frac{1}{10} \times 74 - (0.4)^2\right)$
 $= 4(7.4 - 0.16) = 4 \times 7.24 = 28.96$

Hence $\sigma_x = 5.38$

Correct $\rightarrow C.V = \frac{S.D.}{Mean} = \frac{5.38}{30.8} = 0.1747$

7.	A	$n_1 = 100$	$\bar{x}_1 = \text{Rs. } 460.0$	$\sigma_1 = 50$
	B	$n_2 = 80$	$\bar{x}_2 = \text{Rs. } 490.0$	$\sigma_2 = 40$

- i) Total wage paid by A : $460 \times 100 = \text{Rs. } 46000.00$
 Total wage paid by B : $490 \times 80 = \text{Rs. } 39200.00$
 Factory A pay large amount of weekly wage

ii) C.V. of A $\frac{S.D.}{Mean} = \frac{50}{460} = 0.1087$
 C.V. of B $\frac{S.D.}{Mean} = \frac{40}{490} = 0.0816$

Factory A shows greater variability in the distⁿ of wages as $CV_A > CV_B$.

iii) $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{(n_1 + n_2)} : \text{Simplify}$

$$\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)}}$$

$$d_1 = \bar{x}_1 - \bar{x}, \quad d_2 = \bar{x}_2 - \bar{x}$$

Simplify: